

Single-cell RNA sequencing reveals intra-tumoral heterogeneity of glioblastoma and a pro-tumor subset of tumor-associated macrophages characterized by EZH2 overexpression.

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Glioblastoma (GBM) is a highly heterogeneous disease with poor clinical outcome. To comprehensively dissect molecular landscape of GBM and heterogeneous distribution and potential role of Enhancer of zeste homolog 2 (EZH2) in tumor microenvironment (TME). Single-cell RNA sequencing (scRNA-seq) analysis was performed in GBM samples from 8 patients. Deconvolution analysis, immunofluorescence (IF) microscopy, reverse-transcription quantitative polymerase chain reaction (RT-qPCR), colony formation experiments, and Cell Counting Kit-8 (CCK-8) assays were performed to confirmed the potential role of EZH2 in TME cells. Malignant cells exhibited remarkable heterogeneity in abnormal metabolic patterns. A mesenchymal-2-like (MES2-like) GBM subcluster with glial-immune dual feature was firstly discovered, which were associated with highly activated hallmark pathways, immune evasion associated transcription factor (IRF8), and poor survival. The oncogene, EZH2, was heterogeneously expressed in malignant cells and immune cells consistent with proliferative genes, cell-cycle transcription factors, and similar activated hallmark pathways. In a tumor-associated macrophages (TAMs) subset (macrophage.3), EZH2 was highly expressed with similar changes of transcriptomic dynamics with cell-cycle genes and macrophages M2-phenotype genes. In addition, the subset tightly interacted with malignant cells. Deconvolution analysis showed increased abundance of the subset in GBM compared to low-grade glioma (LGG) and significant association with worse prognosis. Functional verification experiments confirmed the pro-tumor role of TAMs with EZH2 overexpression in GBM. Our study illustrated a MES2-like GBM subcluster characterized by glial-immune dual feature and highlighted the pro-tumor role of a TAMs subset characterized by EZH2 overexpression.

Glioblastomas acquire myeloid-affiliated transcriptional programs via epigenetic immunoediting to elicit immune evasion.

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Glioblastoma multiforme (GBM) is an aggressive brain tumor for which current immunotherapy approaches have been unsuccessful. Here, we explore the mechanisms underlying immune evasion in GBM. By serially transplanting GBM stem cells (GSCs) into immunocompetent hosts, we uncover an acquired capability of GSCs to escape immune clearance by establishing an enhanced immunosuppressive tumor microenvironment. Mechanistically, this is not elicited via genetic selection of tumor subclones, but through an epigenetic immunoediting process wherein stable transcriptional and epigenetic changes in GSCs are enforced following immune attack. These changes launch a myeloid-affiliated transcriptional program, which leads to increased recruitment of tumor-associated macrophages. Furthermore, we identify similar epigenetic and transcriptional signatures in human mesenchymal subtype GSCs. We conclude that epigenetic immunoediting may drive an acquired immune evasion program in the most aggressive mesenchymal GBM subtype by reshaping the tumor immune microenvironment.

T-cell dysfunction in the glioblastoma microenvironment is mediated by myeloid cells

releasing interleukin-10.

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Despite recent advances in cancer immunotherapy, certain tumor types, such as Glioblastomas, are highly resistant due to their tumor microenvironment disabling the anti-tumor immune response. Here we show, by applying an in-silico multidimensional model integrating spatially resolved and single-cell gene expression data of 45,615 immune cells from 12 tumor samples, that a subset of Interleukin-10-releasing HMOX1+ myeloid cells, spatially localizing to mesenchymal-like tumor regions, drive T-cell exhaustion and thus contribute to the immunosuppressive tumor microenvironment. These findings are validated using a human ex-vivo neocortical glioblastoma model inoculated with patient derived peripheral T-cells to simulate the immune compartment. This model recapitulates the dysfunctional transformation of tumor infiltrating T-cells. Inhibition of the JAK/STAT pathway rescues T-cell functionality both in our model and in-vivo, providing further evidence of IL-10 release being an important driving force of tumor immune escape. Our results thus show that integrative modelling of single cell and spatial transcriptomics data is a valuable tool to interrogate the tumor immune microenvironment and might contribute to the development of successful immunotherapies.

Single-Cell Transcriptomics Revealed Subtype-Specific Tumor Immune Microenvironments in Human Glioblastomas.

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Human glioblastoma (GBM), the most aggressive brain tumor, comprises six major subtypes of malignant cells, giving rise to both inter-patient and intra-tumor heterogeneity. The interaction between different tumor subtypes and non-malignant cells to collectively shape a tumor microenvironment has not been systematically characterized. Herein, we sampled the cellular milieu of surgically resected primary tumors from 7 GBM patients using single-cell transcriptome sequencing. A lineage relationship analysis revealed that a neural-progenitor-2-like (NPC2-like) state with high metabolic activity was associated with the tumor cells of origin. Mesenchymal-1-like (MES1-like) and mesenchymal-2-like (MES2-like) tumor cells correlated strongly with immune infiltration and chronic hypoxia niche responses. We identified four subsets of tumor-associated macrophages/microglia (TAMs), among which TAM-1 co-opted both acute and chronic hypoxia-response signatures, implicated in tumor angiogenesis, invasion, and poor prognosis. MES-like GBM cells expressed the highest number of M2-promoting ligands compared to other cellular states while all six states were associated with TAM M2-type polarization and immunosuppression via a set of 10 ligand-receptor signaling pathways. Our results provide new insights into the differential roles of GBM cell subtypes in the tumor immune microenvironment that may be deployed for patient stratification and personalized treatment.

Immunogenetics of glioblastoma: the future of personalized patient management.

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The prognosis of glioblastoma has changed little over the past two decades, with only minor improvements in length of overall survival through the addition of temozolomide (temodal) to standard of care and the recommended use of alternating electric field therapy (optune) to newly diagnosed patients.

In an effort to define novel therapeutic targets across molecularly heterogeneous disease subgroups, researchers have begun to uncover the complex interplay between epigenetics, cell signaling, metabolism, and the immunosuppressive tumor microenvironment. Indeed, IDH mutations are now recognized as a defining differential factor not only influencing global hypermethylation and patient prognosis but also degree of immune infiltration within individual tumors. Likewise, next-generation sequencing has defined subgroup-specific transcriptional profiles that correlate with different mechanisms of immune evasion, including increased PD-L1 and CTLA-4 among mesenchymal tumors. Interestingly, sequencing of the T cell repertoire from numerous patient samples suggests that the correlation between mutational burden and enrichment of tumor-specific peptides may be less convincing than originally suspected. While this raises questions over the efficacy of dendritic cell or tumor-lysate vaccines and CAR-T therapies, these avenues continue to be explored. In addition to these active immunotherapies, inhibitors of molecular hubs with wide reaching effects, including STAT3, IDO, and TGF- β , are now in early-phase clinical trials. With the potential to block intrinsic biological properties of tumor growth and invasion while bolstering the immunogenic profile of the tumor microenvironment, these new targets represent a new direction for GBM therapies. In this review, we show the advances in molecular profiling and immunophenotyping of GBM, which may lead to the development of new personalized therapeutic strategies.

IGFBP2 promotes immunosuppression associated with its mesenchymal induction and Fc γ RIIB phosphorylation in glioblastoma.

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Immunotherapy shows a promise for treating glioblastoma (GBM), the most malignant and immunosuppressive glioma. The mesenchymal phenotype of cancer cells was frequently reported to be associated with their induction of immunosuppression within the cancer microenvironment. Overexpressed insulin-like growth factor binding protein 2 (IGFBP2) promotes GBM cell migration and invasion, and contributes to glioma progression and cancer recurrence and poor survival in GBM. However, whether IGFBP2 can induce immunosuppression in GBM was not reported yet. Thus, the study applied a syngeneic mouse GBM model, human GBM samples, and cancer-immune cell co-culture experiments to investigate the effect of IGFBP2 on GBM exposed immune cells and its association with the mesenchymal induction. We found that IGFBP2 promoted the mesenchymal feature of GBM cells. The inhibition of IGFBP2 relieved immunosuppression by increasing CD8 $^{+}$ T and CD19 $^{+}$ B cells and decreasing CD163 $^{+}$ M2 macrophages. Further, the IGFBP2-promoted immunosuppression was associated with its induction of the mesenchymal feature of GBM cells and the inhibitory phosphorylated Fc γ RIIB of GBM exposed immune cells. Blocking IGFBP2 suppressed tumor growth and improved survival of tumor bearing mice in the mouse GBM model. These findings support the notion that targeting the IGFBP2 may present an effective immunotherapeutic strategy for mesenchymal GBMs.

Dissecting the immunosuppressive tumor microenvironments in Glioblastoma-on-a-Chip for optimized PD-1 immunotherapy.

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Programmed cell death protein-1 (PD-1) checkpoint immunotherapy efficacy remains unpredictable in

glioblastoma (GBM) patients due to the genetic heterogeneity and immunosuppressive tumor microenvironments. Here, we report a microfluidics-based, patient-specific 'GBM-on-a-Chip' microphysiological system to dissect the heterogeneity of immunosuppressive tumor microenvironments and optimize anti-PD-1 immunotherapy for different GBM subtypes. Our clinical and experimental analyses demonstrated that molecularly distinct GBM subtypes have distinct epigenetic and immune signatures that may lead to different immunosuppressive mechanisms. The real-time analysis in GBM-on-a-Chip showed that mesenchymal GBM niche attracted low number of allogeneic CD154+CD8+ T-cells but abundant CD163+ tumor-associated macrophages (TAMs), and expressed elevated PD-1/PD-L1 immune checkpoints and TGF- β 1, IL-10, and CSF-1 cytokines compared to proneural GBM. To enhance PD-1 inhibitor nivolumab efficacy, we co-administered a CSF-1R inhibitor BLZ945 to ablate CD163+ M2-TAMs and strengthened CD154+CD8+ T-cell functionality and GBM apoptosis on-chip. Our ex vivo patient-specific GBM-on-a-Chip provides an avenue for a personalized screening of immunotherapies for GBM patients.

Molecular mechanism by which CDCP1 promotes proneural-mesenchymal transformation in primary glioblastoma.

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Compared with the proneural (PN) subtype of glioblastoma (GBM), the mesenchymal (MES) subtype is more invasive and immune evasive and is closely related to poor prognosis. Here, we used transcriptome data and experimental evidence to indicate that CUB domain-containing protein 1 (CDCP1) is a novel regulator that facilitates the transformation of PN-GBM to MES-GBM. The mRNA expression data of CDCP1 in glioma were collected from the TCGA, CGGA and GEO databases, and in vitro experiments verified CDCP1 expression in glioma tissue samples. Independent prognostic analysis revealed the correlation of the CDCP1 expression level and patient survival. Bioinformatics analysis and experiments verified the biological function of CDCP1. Multivariate proportional hazards models and a PPI network were used to select key genes. A prognostic risk model for predicting the survival of glioma patients was constructed based on the selected genes. The results showed that the expression of CDCP1 increased with increasing tumor grade and that the overexpression of CDCP1 correlated with a poor prognosis. CDCP1 was highly expressed in MES-GBM but weakly expressed in PN-GBM. The risk model (considering CDCP1 combined with CD44 and ITGAM expression) could represent a tool for predicting survival and prognosis in glioma patients. Our study indicates that CDCP1 plays an important role in facilitating the transformation of PN-GBM to MES-GBM.

Profiling pro-neural to mesenchymal transition identifies a lncRNA signature in glioma.

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Molecular classification has laid the framework for exploring glioma biology and treatment strategies. Pro-neural to mesenchymal transition (PMT) of glioma is known to be associated with aggressive phenotypes, unfavorable prognosis, and treatment resistance. Recent studies have highlighted that long non-coding RNAs (lncRNAs) are key mediators in cancer mesenchymal transition. However, the relationship between lncRNAs and PMT in glioma has not been systematically investigated. Gene expression profiles from The Cancer Genome Atlas (TCGA), the Chinese Glioma Genome Atlas (CGGA),

GSE16011, and Rembrandt with available clinical and genomic information were used for analyses. Bioinformatics methods such as weighted gene co-expression network analysis (WGCNA), gene set enrichment analysis (GSEA), Cox analysis, and least absolute shrinkage and selection operator (LASSO) analysis were performed. According to PMT scores, we confirmed that PMT status was positively associated with risky behaviors and poor prognosis in glioma. The 149 PMT-related lncRNAs were identified by WGCNA analysis, among which 10 (LINC01057, TP73-AS1, AP000695.4, LINC01503, CRNDE, OSMR-AS1, SNHG18, AC145343.2, RP11-25K21.6, RP11-38L15.2) with significant prognostic value were further screened to construct a PMT-related lncRNA risk signature, which could divide cases into two groups with distinct prognoses. Multivariate Cox regression analyses indicated that the signature was an independent prognostic factor for high-grade glioma. High-risk cases were more likely to be classified as the mesenchymal subtype, which confers enhanced immunosuppressive status by recruiting macrophages, neutrophils, and regulatory T cells. Moreover, six lncRNAs of the signature could act as competing endogenous RNAs to promote PMT in glioblastoma. We profiled PMT status in glioma and established a PMT-related 10-lncRNA signature for glioma that could independently predict glioma survival and trigger PMT, which enhanced immunosuppression.

Temporal change of DNA methylation subclasses between matched newly diagnosed and recurrent glioblastoma.

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The longitudinal transition of phenotypes is pivotal in glioblastoma treatment resistance and DNA methylation emerged as an important tool for classifying glioblastoma phenotypes. We aimed to characterize DNA methylation subclass heterogeneity during progression and assess its clinical impact. Matched tissues from 47 glioblastoma patients were subjected to DNA methylation profiling, including CpG-site alterations, tissue and serum deconvolution, mass spectrometry, and immunoassay. Effects of clinical characteristics on temporal changes and outcomes were studied. Among 47 patients, 8 (17.0%) had non-matching classifications at recurrence. In the remaining 39 cases, 28.2% showed dominant DNA methylation subclass transitions, with 72.7% being a mesenchymal subclass. In general, glioblastomas with a subclass transition showed upregulated metabolic processes. Newly diagnosed glioblastomas with mesenchymal transition displayed increased stem cell-like states and decreased immune components at diagnosis and exhibited elevated immune signatures and cytokine levels in serum. In contrast, tissue of recurrent glioblastomas with mesenchymal transition showed increased immune components but decreased stem cell-like states. Survival analyses revealed comparable outcomes for patients with and without subclass transitions. This study demonstrates a temporal heterogeneity of DNA methylation subclasses in 28.2% of glioblastomas, not impacting patient survival. Changes in cell state composition associated with subclass transition may be crucial for recurrent glioblastoma targeted therapies.